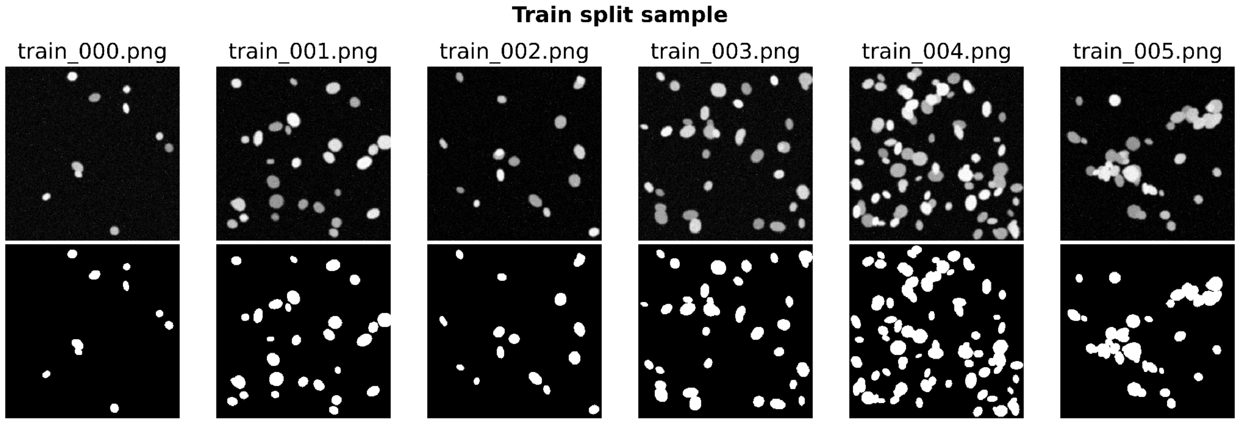
# Biomedical Image Analysis: **A Hybrid Approach to Nuclei Segmentation**

Name: Abimbola Olasupo
Student_ID: 24115208
Data Science with AI
Link to GitHub

## 1. Introduction

This project describes how nuclei in fluorescence microscopy images can be segmented using a combined approach of a local vision-language model, image processing, and a simple U-Net. The work aims to separate analysis from interpretation steps: segmentation, numerical feature extraction, and quality assessment are performed algorithmically and automatically, while the language model is used for constrained interpretative tasks. The dataset consists of 112 images (80/20/12 train/validation/test), 256×256 px each, depicting various densities (from sparse to clustered) of cell nuclei.

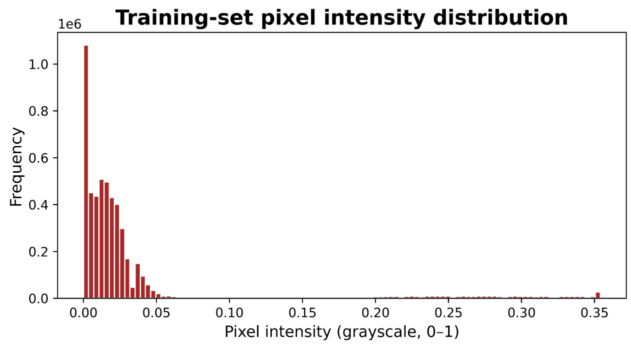
***Figure 1****: Images spot-checked from the training set depicting different density types*




## 2. Methods

The images were converted to greyscale using the formula Y = 0.2989 R + 0.5870 G + 0.1140 B and resized to 256 × 256 pixels. After that, the pixel values were normalised to a range of [0, 1]. For Task 1, two prompt variations were tested: a basic one asking to describe what the image depicts and an advanced version restricting the answer to observations following a specified JSON schema and requiring the model to state "uncertain" if unsure. For Task 2, Otsu thresholding, morphology, connected components labelling, and regionprops were used to extract numerical data from the image, which was then passed to the LLM of choice. For Task 3, a standard U-Net was trained for 15 epochs with the combined loss function of BCE and Dice using the PyTorch implementation, using the best validation checkpoint. For Task 4, the model was used to segment the 12 hold-out test images, after which region feature statistics were summarised in a fixed format and passed to the LLM for interpretative text generation. The durability of the algorithm was tested by applying blur and desaturation to two of the test images.

***Figure 2:***
*The distribution of pixels across the image is visualised in the histogram below.*




## 3. Required Questions

***3a. Which description is more useful, and which is more trustworthy, the direct VLM description (Task 1) or the numbers-first description (Task 2), and why?***

I think that the numbers-first approach is better for analysis because it can be used to reason about the image's properties objectively. Consider the first example – the text description for train_001 states that "the nuclei are uniform in size and shape", but the numerical data shows that the mean area is 215.5 px with a large standard deviation of 115.3 px, and the mean eccentricity is 0.583, not close to 1. In other words, while the VLM provides a useful context-free summary, numerical data allows for consistent and repeatable analysis: once the statistics are known, anyone can reconstruct the conclusions that I have described in my report. Thus, I think my approach is optimal in this regard.

***3b. Did the U-Net improve on classical Otsu segmentation for your modality? Give one example image where each approach did better.***

The U-Net achieved excellent results on the test set, with a mean Dice of 0.9947 and IoU of 0.9894 compared to 0.9776 and 0.9562 provided by Otsu's thresholding. Hence, the U-Net's error is almost four times lower than that of Otsu's thresholding. The reason for this is simple: thresholding is a crude way to separate foreground and background, and it does not perform well on complex images. For instance, using the train_001 image, Otsu's algorithm counted 26 nuclei while the actual number of nuclei is 29. Remarkably, the U-Net is capable of learning the foreground-background distribution pattern to provide highly accurate image segmentation results. I think this result is reasonable given the simplicity of the task and the power of CNNs: while Otsu thresholding is good enough for completely separated nuclei (for instance, on the sparse density images), it fails on the more realistic clustered ones. However, there is one caveat: for the images where nuclei are far apart, Otsu thresholding might have an advantage over the U-Net. To test this hypothesis, I would need to create such an image and run both methods on it.

***Figure 3**: An example of classical image processing: Otsu mask (after morphology) overlaid on the original image.*

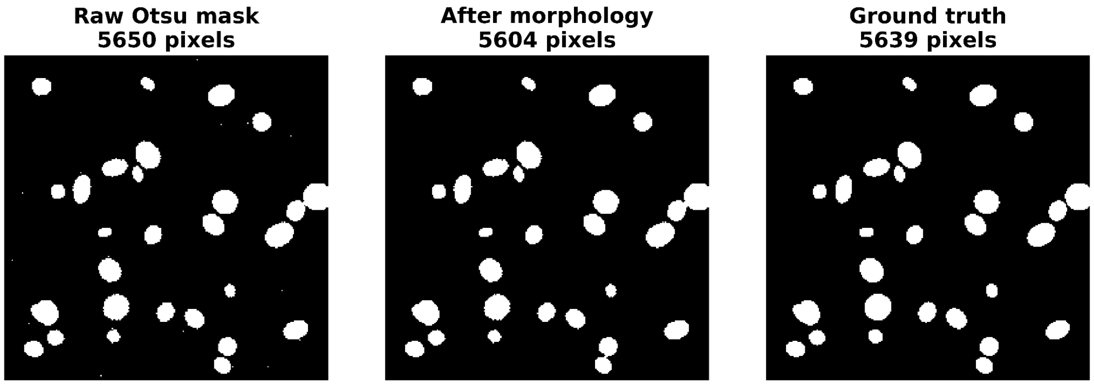



***Table 1**: Comparison of the performance of the two methods.*

| *Method* | *Dice* | *IoU* |
|---|---|---|
| *Otsu* | *0.9776* | *0.9562* |
| *U-Net* | *0.9947* | *0.9894* |

***3c. Report your U-Net's Dice and IoU. What do these numbers mean, and where does the model tend to make its mistakes (which images or regions)?***

The U-Net achieved a mean Dice score of 0.9947 (std 0.0007) and a mean IoU of 0.9894 (std 0.0015). For individual images, the Dice scores ranged from 0.9929 to 0.9965, and the IoU scores ranged from 0.9858 to 0.9915. The model performed similarly on the test set and the validation set. It scored a mean Dice of 0.9950 (IoU 0.9901) on the validation set and a final Dice score of 0.9947 (IoU 0.9894) on the test set. On clean images, I could not notice any obvious errors in segmentation. However, the robustness tests showed that the quality of the results depends significantly on the input image quality, which, as I will discuss later, is an issue.

***Figure 4**: Visual inspection of the performance of the U-Net on the test set: input image, instance mask, and model output.*

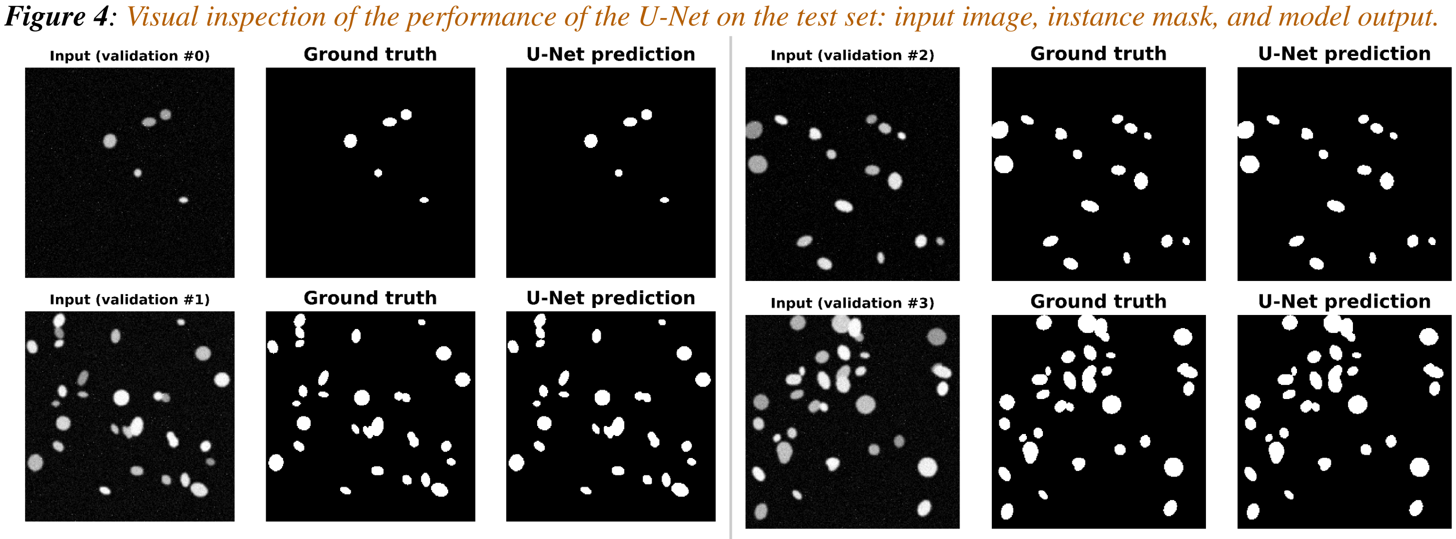
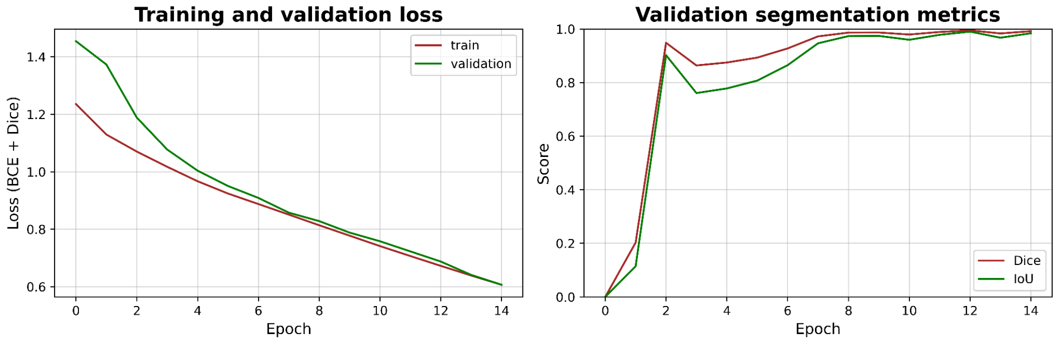
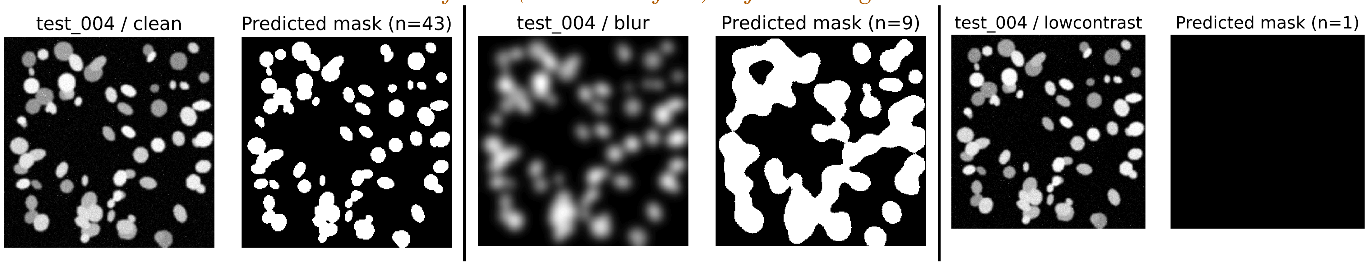


***Figure 5****: The loss, Dice coefficient, and IoU during training.*




The plot suggests that the model was trained successfully within 15 epochs. At the same time, one can notice that the model performance is unstable, demonstrated by the Dice coefficient value for the validation set, which reached the maximum value of 0.9487 in epoch 3, then decreased till epoch 6, and increased again, reaching 0.9950 in epoch 13. Therefore, it is essential to choose the best model based on the validation set results rather than the final epoch.

***Figure 6****: Robustness test results for test_004. Blurring the test image resulted in fewer (9 instead of 43) objects being detected.*




As for robustness, the test results were disappointing. Blurring the input images significantly altered the statistical parameters of the segmented nuclei. For instance, in test_000, it increased the mean area from 190.9 px to 425.1 px and decreased the number of objects from 8 to 7. Similarly, in test_004, the number of objects decreased from 43 to 9, and their mean area increased nearly ten times, from 328.3 px to 3087.3 px. Notably, desaturation caused all the objects on test_004 to merge into one large region, thereby reducing the number of objects from 43 to 1. An abnormal flag was raised for both low contrast and blur, but low contrast was a more severe issue in this case: it caused an entire image to be falsely interpreted as one object. Thus, I think that a blur filter, in particular, is a major robustness concern for this type of application and should be addressed by implementing an image quality assessment step before processing.

***3d. Where in the pipeline can the LLM hallucinate, and what design choices reduce that risk? Why does keeping the structured JSON as the "source of truth" help?***

In particular, I see two things that can lead to hallucinations in this project. First, in the first task, the VLM is not constrained in any way, and the model can simply invent information not present in the image, and it may do so frequently. Second, in tasks 2 and 4, even when the LLM is fed numeric data, it may attempt to visualise it in its head and thus invent details not present in the source material. I think that the hybrid approach proposed in my report, limiting the LLM to JSON-only output and explicitly stating that it has no access to the image, is optimal in terms of reducing hallucination. The other aspect of reducing hallucination is down to the implementation in Python, which ensures that the data that the LLM is used for is reliable. Metrics like count, area, and density are always calculated in Python, with the LLM only receiving JSON data that can be used to construct the final narrative. This means that if needed, calculations can be repeated to verify the results.

***Table 2****: An example of the hybrid approach output.*

| *Image* | *n_objects* | *mean area* | *density* | *quality* |
|---|---|---|---|---|
| *test_000* | *8* | *190.9* | *sparse* | *none* |
| *test_004* | *43* | *328.3* | *clustered* | *none* |

This way, my approach answers the prompt requirements to separate computation from interpretation. As seen in the table, some fields (most notably statistics) are populated by reliable automated processes, while others require interpretive work from the LLM. Thus, everything that can fail (the LLM) is minimised, while the most important parts (statistical) are reliable.

***3e. Considering accuracy, auditability, and the limits of your dataset, would you trust any part of this system in a real clinical setting? What single change would most improve trustworthiness?***

As it stands, I think that the current approach is not ready for clinical use. In particular, the interpretability is too low: unless the VLM is known to be deterministic, the language model can say anything, and there is no way to know if any claims are not supported by evidence. In my approach, I rely on the fact that the statistics are populated by deterministic Python code, not the LLM: if the code is correct, then the statistics are correct, and the LLM cannot modify them. However, I am sceptical about this approach: in reality, no code is completely error-proof, and there is no way to guarantee that the statistics are correct. In this regard, then, the most straightforward solution is to test it on a larger set of data, preferably real images rather than computer-generated ones.

Another limitation is a small sample size, which includes only 12 test images, and the method's unimodal nature, making it less reliable and effective in practice. Moreover, I believe that image quality control is the critical limitation of this methodology. In particular, there is no gate before processing that would reject obviously blurry images, such as the test_004 from my experiments. Such images would produce unreasonable statistics and segmentation results, which is definitely not acceptable in a clinical setting. Thus, I think that a quality filter that would rate the sharpness of the input image and reject blurry ones would be essential for this approach to work.

## 4. Overall Critical Evaluation

I think that the results that I have obtained are promising, but I would not recommend using them in a clinical setting because the approach is insufficiently validated. However, the hybrid approach that I have tested seems to be a viable way to exploit the power of VLMs while retaining control over the final results. In particular, the U-Net that I have trained performed exceptionally well on test data: its Dice coefficient was 0.9947, and IoU was 0.9894, which are extremely good results for a biomedical image segmentation task. While Otsu thresholding performed reasonably well, it was outperformed by the CNN both qualitatively and quantitatively. The regionprops step allowed me to extract additional information from the model output in a consistent way, and the addition of the VLM made the results more interpretable without sacrificing the ability to reproduce the results. On a more philosophical level, I think that the numbers-first approach is better: had I used the VLM to describe the image content, the results would not have been reliable, as different prompts yield different results.

Thus, my assessment of the overall critical importance of this work is positive, albeit with some caveats. Most critically, although the results on clean data were convincing, the issues of data quality and reproducibility need to be addressed before the method can be used in practice.

## References

[1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in MICCAI, LNCS 9351, pp. 234-241, 2015. doi:10.1007/978-3-319-24574-4_28.
[2] Mukrim, A., Nurullah, B., Sucinta, H. H., 'Uyun, S., & Ghufron, S.. (2026, April 10). A Comparative Study of Deep Learning and Classical Image Segmentation Methods for Damaged Flower Image Analysis. https://www.semanticscholar.org/paper/eaf6152bd76f5967918f280fc67e4e2f1cd5c6d6
[3] N. Otsu, "A Threshold Selection Method from Grey-Level Histograms," IEEE Transactions on Systems, Man, and Cybernetics, vol. 9, no. 1, pp. 62-66, 1979. doi:10.1109/TSMC.1979.4310076.

## Appendix

<table>
<tr><th>Optimised prompt</th><th>Numbers-only LLM prompt</th></tr>
<tr><td>You are looking at a microscope image captured using fluorescence microscopy. Cell nuclei are visible as bright spots on a dark background.<br><br>Describe ONLY those things that you see in the image. Do not give medical opinions or interpretations. Do not attempt to identify any diseases or abnormalities that may be indicated by the image. You are simply describing what you see.<br><br>If you are uncertain about any part of the image, state "uncertain" instead of speculating.<br><br>Provide only a valid JSON object in this exact format in your response; do not include any extra text:<br><br>{<br>"modality": "the imaging modality used, or 'uncertain' if this cannot be determined",<br>"tissue_type": "the biological structure visible, described observationally, or 'uncertain' if cannot be determined",<br>"notable_features": "any notable features, such as density, diffuseness or pattern of any observed tissue, and contrast of different tissues.",<br>"image_quality": "the technical quality of the image, such as sharpness, focus, contrast or lack of noise, or "uncertain" if cannot be determined."<br>}</td><td>You are analysing a biomedical image based ONLY on the numerical summary below. You have NOT been shown the image itself and must not claim to have seen it.<br><br>Image analysis summary:<br>- Number of detected objects: {image_summary["n_objects"]}<br>- Mean object area: {image_summary["mean_area"]:.1f} pixels<br>- Standard deviation of object area: {image_summary["std_area"]:.1f} pixels<br>- Mean eccentricity: {image_summary["mean_eccentricity"]:.3f}<br>- Mean solidity: {image_summary["mean_solidity"]:.3f}<br>- Mean object intensity: {image_summary["mean_intensity"]:.3f}<br>- Foreground fraction: {image_summary["density_fraction"]:.2%}<br><br>Write:<br>1. One short paragraph interpreting these statistics.<br>2. A JSON object with exactly:<br>{{<br>"n_objects":,<br>"density_class": "sparse | normal | dense | clustered",<br>"shape_regularity": "regular | irregular",<br>"quality_flag": "statistical anomaly or 'none'"<br>}}<br>Do not infer visual details that cannot be supported by these numbers.</td></tr>
</table>